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Fake News Detection using Machine Learning and Deep Learning Models

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Abstract

This dissertation focuses on the identification of fake news using machine learning and deep learning techniques. First, the dataset is cleaned and preprocessed, and the features are engineered. The basic classification algorithms employed are Naive Bayes, Support Vector Machine (SVM), Random Forest, and XGBoost, and the deep learning algorithm is a Bidirectional Long Short-Term Memory (BiLSTM) model. To further improve the detection performance, sentiment analysis is also implemented into the models. The performance of these models is measured accordingly in terms of accuracy, precision, recall and F1-score between real and fake news articles. The results show that the BiLSTM model has higher accuracy in comparison to the machine learning algorithms such as XGBoost, which is a testament to the fact that deep learning techniques yield better results in terms of capturing the intricate patterns and semantics of fake news.

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Introduction

1.1 Background

Definition and Characteristics

Low-quality news that purposefully contains misleading information to deceive readers is referred to as fake news. It consists of political statements, rumours, bogus reviews, and deceptive information. "News articles that are intentionally and verifiably false, and could mislead readers" is how Stanford University defines fake news. According to Wikipedia, false news is "a type of yellow journalism or propaganda that consists of deliberate misinformation or hoaxes spread via traditional print and broadcast news media or online social media" [3]. Related terms include deceptive news, false news, satire news, disinformation, misinformation, cherry-picking, clickbait, and rumours [4]. Social media amplifies the reach and impact of fake news due to its low cost, easy access, and rapid dissemination capabilities [1]. Misinformation (false or misleading information) and disinformation (false information disseminated to deceive others) are synonyms for fake news. Fake news can moreover be found on subjects like stock values, diet, and vaccinations, although it is becoming more common in political contexts lately. It is poisonous because it preys on and damages the reputation of mainstream media sources [2].

Historical Context

The term "fake news" dates back to the 1890s, but the phenomenon has roots as early as Johannes Gutenberg's development of the printing press in 1439. Historic examples include the "Great Moon Hoax" (1835) and the "blood libel" story from Trent, Italy (1475) [6], but social media has made it a more powerful force, challenging traditional journalistic norms [1]. Journalism objectivity and balance were founded in response to propaganda during World War I and the developing corporate control of America's Communication in the 1920s. These norms were supported by the print and broadcast media that controlled entry into the field in the twentieth century, but the Internet brought new entrants who rejected these norms and undercut traditional news providers[2]. While the occurrence of fake news was given a lot of attention and was held responsible for raising political polarization and manipulation during the 2016 U.S. presidential elections[3], both the Oxford and Collins dictionaries gave the title of 'term of the year' to the concept of 'fake news' in 2016 and 2017 respectively. The widespread of new means of communication has facilitated the swift creation of fake news and their circulation which has raised the problem of differentiating between real and fake news [6].

Psychological and Social Foundations

Individuals typically lack critical thinking skills and commit mistakes like confirmation bias and naive realism, which are the tendency to accept information that confirms one's judgment as true. According to this study, users are willing to spread false information to get social approval from their groups, which is consistent with social identity theories and normative influence. Therefore, news consumption and distribution can be understood in the context of game theory, in which participants want to maximize their pleasure by making the most use of the resources at their disposal. This is the scenario when the personal advantages of creating and consuming fake news influence consumers and producers, respectively, to cater to short-term interests [1]. Fake news can overwhelm the set perspectives, concerns, and choices affecting interactions between real news. It can lead users, especially the juvenile, the teenagers and the aged people, into misleading information. Another case took place in South America, when, after the

earthquake in Chile in 2010, people's panic was amplified due to the rumours, spread through the social networking site Twitter. Miseducative stories like a fake news story of CNN's improved ratings or that Stephen Hawking was a proponent of aliens are evidence of how deep fake news goes [3]. Cognitive biases and pro-sociality of humans affect fake news consumption and being fake news. Social factors such as repetition (validity effect), providing feedback to your previous judgment (confirmation bias), desire for something to be true (desirability bias), and outsiders pressure are some of the ways that make a person believe and spread fake news [4].

Prevalence and Impact

According to the studies, it was hypothesised that in a significant event like the 2016 United States presidential election, there were more than one million tweets concerning a single fake news article. This means that the incidence of fake news is evidently high. The method by which fake news spreads information misleads the general public and hinders them from distinguishing between real and genuine information. It can also disturb the equilibrium of news authenticity types and force the news consumers to make wrong perceptions [1]. In the investigation of one research [5], in the month before the 2016 election one to three pieces of fake news from reliable sources were seen by the average American. The results showed that the propagation rate of falsehoods is higher than that of the truth across different platforms including the Social networking site Twitter. Many of these are bots or automated accounts, making them one of the main sources of fake news. About 60 million Facebook accounts, 9-15% of daily active Twitter accounts approximately estimates suggest that accounts may be bots [2]. Fake news poses a clear and inherent danger to public trust, democracy, and justice. Some of the effects may have included influencing important political events such as the Brexit referendum or the 2016 United States presidential election and contributing to the loss of public trust in governments [4]. Fake news can harm economies, for instance, when a report that President Obama was injured leading to his death, results in a loss of about \$130 billion in the stock market. It can also be for the target group, political, or financial (i.e. clickbait stories, propaganda). When commercial messages are inserted into public relations and advertising using journalistic appeal, these may also be classified as false news [6].

1.2 Motivation

The propagation of fake news is greatly aided by the existence of social bots, cyborg users, and trolls on social media sites. Large-scale online conversation manipulation and distortion are possible with these accounts. Social media fake news generates large amounts of noisy, unstructured, incomplete, and massive data, making detection difficult. The challenge is exacerbated by the quick appearance of novel, time-sensitive occurrences, which may not be confirmed by current information systems [1]. Online social media's fast extension has made it simpler for data to spread quickly, which has made it a popular venue for the propagation of false information. Websites such as wtoe5news.com and denverguardian.com disseminate propaganda, hoaxes, and misinformation for monetary or political advantage, publishing fake news that looks like it comes from real news sources. Real-time identification of fake news is difficult since it frequently leverages current events as a focal point to garner attention [3]. In this respect, various social media platforms have tremendously augmented the production and distribution of fake news. Social media removes the physical barrier of news sharing and offers diverse environments to share and discuss news, which leads to the circulation of the echo-chamber effect, which means the over-sharing of biased information [4]. Fake news has the negative impact of despising journalism and shifting people's views, which can hinder democratic procedures. Media literacy, development of new fact-checking products, and setting up proper technical, financial, and appropriate legal barriers against fake news distribution are few fixes to it [6].

Detecting fake news needs more than analysing the content; it involves using auxiliary information such as user social engagements, knowledge bases, and network interactions. There is a need for more advanced methods to differentiate credible users, extract useful features from social media posts, and exploit network interactions for better fake news detection [1]. The massive volume and variety of fake news make detection difficult. Fake news includes not only misleading political statements but also fake reviews, advertisements, and conspiracy theories. Detecting fake news is challenging due to the need for real-time analysis and the dynamic, diverse nature of online content. Manual fact-checking is time-consuming and limited in scope [3]. Most current fact-checking methods are based on manual identification of fake information,

but because of the high flow of information on the Internet, there is a need for auto frameworks. Preconditions for detection are the credibility of authors, spreading patterns and other characteristics besides the text analysis. Possible areas related to the research topic are unsupervised learning schemes, one-class classification schemes and real-time detection schemes. The aim is to come up with positive feedback mechanisms that can easily screen fake news for rejection [3].

Thus, it is imperative to look for methods that are automatic, scalable, and accurate in real-time false news detection. When combined with feature engineering, representation learning and advanced machine learning, it is possible to build intelligent systems capable of discovering fake news from the text, users and distribution patterns. These models are cost-efficient and easily customisable for combating fake news and defending online information systems because they can be trained on large amounts of data, revised periodically and effectively used in practical scenarios.

1.3 Research Objectives

The objective of this research is to create an algorithm that uses a variety of machine-learning approaches to detect fake news. The goal of the algorithm is to train it to recognize traits and trends in the data that can aid in differentiating between legitimate and fraudulent news. The algorithm will be able to classify fake news using fresh data once it has been trained using this information. The following questions are the main focus of this research.

- 1. Which machine learning model is most effective for predicting fake news and what metrics are most useful for evaluating its performance?**

The research aims to implement various Machine Learning models: Naive Bayes, Support Vector Machine; Ensemble methods: Random Forest, XGBoost; Deep Learning model: BiLSTM (Bidirectional Long Short-Term Memory). These models will be trained based on a fake news dataset and evaluated using relevant performance metrics.

- 2. Will the deep learning model outperform machine learning ensemble methods in terms of accuracy and efficiency in detecting fake news?**

The capacity of deep learning models to automatically extract and learn complex patterns from huge datasets generally leads to superior performance for fake news identification when compared to traditional machine learning ensemble methods. The aim is to compare the performance of both methods to effectively detect fake news.

3. In what ways does sentiment analysis affect the machine learning model's overall efficacy in identifying fake news?

A significant amount of research is still being done on how people are emotionally duped by misleading information for both social and personal gain. The effectiveness of a fake news detection algorithm is investigated in two scenarios in this study: one that uses text-based features and the other that just uses emotive features. It assesses these examples with the same performance standards and goals to determine whether or not sentiment analysis enhances the model's potential.

1.4 Dissertation Outline

The foundation of the entire report is the history of spotting fake news. The next portion of the research presents the full analysis of the study's findings. To comprehend the current body of information and hypotheses, Chapter 2 explores the significance of performing a literature review. An overview of the research methodologies used in the study of identifying false news is given in Chapter 3. It also describes the format and prerequisites required to conduct this kind of research. The step-by-step development process, from preliminary analysis to ultimate implementation, is demonstrated in Chapter 4. An analysis and discussion of the research study's findings are given in Chapter 5. A thorough review and final analysis of the major discoveries made during the investigation are given in Chapter 6. It also discusses the potential for additional study and investigation in the future.

Literature Review

2.1 Early Approaches: Traditional Methods

The initial approaches included domain expertise, rule-based systems, and machine learning to combat fake news. Several predetermined rules and/or heuristics were specified in these methods, taking into account linguistic and writing quirks or specific patterns characteristic of fake news [9, 10]. For example, Rubin *et al.* [11] established a paradigm for classifying false news into three categories: "mass" fakes, or hoaxes, "trifling fakes," and "gross" fakes, or those that genuinely mislead readers. They were able to classify a number of characteristics in the newspaper articles, such as tone, punctuation, and terms that might be used to differentiate between genuine and fraudulent news.

Although these initial approaches were helpful in some ways, they were not very effective at scale due to their inability to constantly update to new forms of fake news, and the reliance on manual labour in updating and maintaining the sets of rules. Also, they frequently described difficulties in distinguishing the characteristics of fake news which may differ depending on the domain.

2.2 Machine Learning approaches

As the application of machine learning (ML) advances, scholars began to attempt to use supervised models for the identification of fake news. These approaches generally included feature extraction from the textual content, user-related information, and network features that used supervised labelled datasets for training classifiers [12, 1], which are explained as follows.

- **Content-based Features:** Several works have aimed at obtaining content-based features from the textual information of news articles or posts in social media. These features can be n-grams that are sequences of n words, linguistic patterns, stylometric features such as readability and sentiment and writing style and topic models [13, 14]. Other researchers have also focused on the psychological and deception indicators which include: hedges, pronouns, and emotion/cognition combinations [15, 16].
- **User-based Features:** Apart from content-based features, user-based features have been utilized to consider the credibility and activity of the source or author. Some of these aspects may involve the user's details, social network attributes such as the number of followers, influence metrics and the posting history of the user [17, 18].
- **Network-based Features:** Communication networks and the peculiarities of their dissemination can be successfully used as a basis for understanding the spread of fake news. Others are temporal and structural properties of information cascades on the web which can be utilized for fake news detection [19].

After extracting related features, we develop different supervised ML models to be trained on labelled datasets for the classification of fake news. Logistic regression, support vector machines (SVM), decision tree, random forest, and gradient boosting machines are popular models in this area [20, 21]. Ensemble methods have been employed to enhance the overall effectiveness and the robustness of fake news detection systems, unifying different base models [22].

Feature engineering solutions have been mostly successful, but the process takes time, often depends on the field and does not always handle changes in fake news well.

Also, the exclusive use of hand-crafted features may hinder the generality of the ML models to other types of fake news not sampled in the model's training set.

2.3 Deep Learning approaches

In an attempt to overcome feature engineering limitations and embrace representation learning, an increasing number of deep learning (DL) models have been applied to identify fake news. These models can learn the representations directly from raw data without much human effort, thus there is a possibility of learning very complex patterns and contextual features which might be hard to code manually.

- **Recurrent Neural Networks (RNNs) and Variants:** RNN classifiers like Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) are particularly suitable for sequence modelling and text categorization problems including fake news detection [23, 24]. Such models can include features such as long-range dependencies and contextual information for textual data; hence, applying them to articles such as news articles or posts on social media networks is possible.
- **Convolutional Neural Networks (CNNs):** CNNs, outlined for computer vision tasks, have also been used in text classification scenarios such as fake news detection. CNNs are capable of learning the hierarchical representations and can capture local patterns of text that make them useful for learning the n-gram features and extracting the relevant semantic information [25, 26].
- **Attention Mechanisms:** Most of the models contain important characteristics of RNN and CNN wherein attention mechanisms have been incorporated to direct the models to the important parts of the input text for fake news classification. These mechanisms can aid the models more in understanding the contextual information and flag salient features [24, 27].
- **Multimodal and Hybrid Approaches:** Since the fake news can be in text, image, video, etc. the researchers focused on the multimodal system that incorporates information from different modalities. These models can apply textual and visual features, propagation patterns, as well as user behaviours in intensifying the fake news detection efficiency [28, 29].

2.4 Datasets and Evaluation Metrics

A crucial element of research on fake news detection is the accessibility of high-quality datasets for ML and DL model evaluation and training. The following datasets have been carefully chosen and often used in the literature:

The **LIAR dataset** [30] comprises brief statements that are categorized according to their degree of veracity, with labels ranging from "true" to "pants on fire," which denotes utter falsity. It was made by personally verifying statements made by well-known people.

The **FakeNewsNet** [31] dataset includes news content, social context (user engagements), and spatiotemporal information, providing a comprehensive resource for studying fake news on social media.

The **PHEME dataset** [28] is focused on rumour detection, and contains conversational threads from various social media platforms, annotated for their veracity and support for different stances.

The **NELA-GT-2018** [32] is a huge dataset of news articles, web pages, and metadata, with labels for different types of misinformation and disinformation.

Besides these datasets, there are some attempts made to build synthetic/semi-synthetic datasets through data augmentation, crowdsourcing and generative adversarial networks (GANs) [33].

Specifically, some performance measures which are often used in false news detection research include accuracy, precision, recall, F1 score, and AUC-ROC. However, in comparing the effectiveness of basic ML and advanced DL models concerning the detection of fake news, investigators have also stressed the need to consider extra factors such as the interpretability of the models, their resilience, and the generalizability of their application [4, 34].

2.5 Research Gaps and Future Directions

Even while ML and DL models have made great strides in the detection of fake news, there are still several research gaps and restrictions.

- Handling Evolving Forms of Fake News: False information is very dynamic and

can transform rapidly, so, the use of the ML and DL models may be very lagging. To solve this problem, there must be constant changes to the model and adaptation measures [35].

- **Limited and Biased Data:** A lot of the existing datasets for fake news detection are small, domain-specific or may become inherently biased, and therefore a large amount of the trained models may not be generalizable. Interpretation: To enhance the performance of the models, there is a necessity for larger, more diversified, and less prejudiced datasets [36].
- **Language and Domain Dependencies:** English has been the most studied language for sources and specific areas (e.g. politics, news), while fake news can exist differently in different languages and areas. It is important to note that finding ways to address cross-lingual and cross-domain misinformation detection is an open issue[36].
- **Interpretability and Explainability:** Deep learning models have exhibited high accuracy but the limitation is that their patterns are not easily derived and in most cases difficult to explain. Explaining black box models is vital for obtaining people's trust and providing them with the option to monitor the results automatically [37].
- **Multimodal Fake News Detection:** Since fake news media messages can be text messages, pictures, videos, etc., we are observing the rising demand for multimodal approaches capable of combining the information received from various sources [38].
- **Adversarial Attacks and Robustness:** The fake news detection systems are easily attacked by adversarial attacks, where attackers aim at deceiving and escaping from the model. Therefore, designing such strong and secure models that may not be vulnerable to such attacks is one of the contemporary research priorities [33, 39].
- **Real-time and Scalable Detection:** Since fake news are also shared through social media platforms, it is crucial to detect them in real-time and at a large scale to be able to process the required amount of data and notify about the potential presence

of such news. To solve this issue it is necessary to apply efficient strategies aimed at model deployment and its optimisation [40].

- **Integration with Fact-checking and Content Moderation:** Some detection models can find out potential misinformation however, they require fact-checking employee interference as well as content moderation. It will require the formulation of strategies for interaction between humans and artificial intelligence and designing interfaces that will suit the use of AI in practical applications [41].
- **Ethical Considerations and Potential Misuse:** Some of the ethical considerations are as follows; Privacy and potential discrimination, thus fake news detecting systems are also capable of causing prejudice based on race, gender or other aspects, Misuse of fake news detecting systems, by authoritarian regimes or by any other person with ill intentions. These issues need to be rectified through standard research practices and regulating policies and guidelines [42].

Nevertheless, several challenges remain in the detection of fake news using ML and DL models, yet the research on this topic is progressively growing due to the increasing awareness of the need to curb the spread of fake news to enhance the credibility of information on social media platforms. Multidisciplinary approaches considering the researchers from different fields like natural language processing, computer vision, social network analysis and human-computer interfaces will become essential for managing such challenges as well as finding systemic solutions.

Moreover, because the advancement in technology and new social media platforms are on the increase, the strategies for detecting fake news should always be updated based on the newest trends and new forms of fake news. Collaborative continuation of research work as well as proper utilization of development and deployment procedures can help create more secure and reliable arenas of information available through Internet media.

Methodology

This chapter gives an overview of the machine learning and deep learning approaches used for conducting effective fake news detection. It also describes the dataset details, performance metrics and technological requirements used to carry out this research.

3.1 Machine Learning Approach

The flowchart shown in Figure 3.1 outlines the steps used to carry out fake news detection using Machine Learning models.

3.1.1 Naive Bayes

Naive Bayes is an algorithm of the category of probabilistic classifiers derived from Bayes theorem of conditional probabilities under the assumption of attribute independencies [43]. Thus it is useful in text classification, identification of spam mail, and determination of sentiment among others because of its efficiency particularly while dealing with big data with high dimensions.

The Bayes theorem forms the basis of Naive Bayes:

$$P(C | X) = \frac{P(X | C) \cdot P(C)}{P(X)}, \quad (3.1)$$

where:

$P(C|X)$ is the posterior probability of class C given predictor X ,

$P(C)$ is the prior probability of class C ,

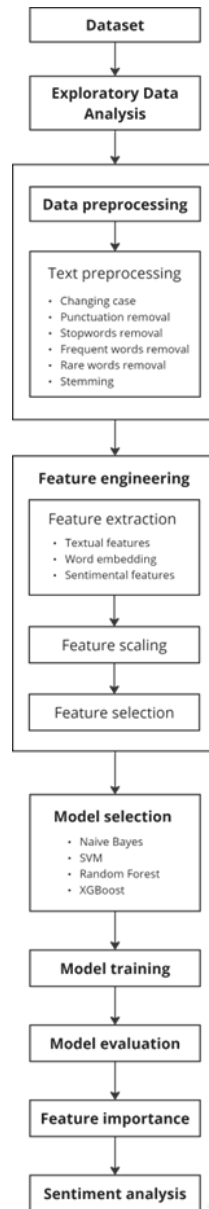


Figure 3.1: Machine Learning methodology.

$P(X|C)$ is the likelihood of X given class C ,

$P(X)$ is the prior probability of X .

Given the class, the "naive" assumption is that features are conditionally independent:

$$P(X | C) = P(X_1 | C) \cdot P(X_2 | C) \cdot \dots \cdot P(X_n | C) \quad (3.2)$$

Not to mention that big feature sets are conceivable, and this simplification enables computing. Naive Bayes comes in three primary varieties [44]:

1. Gaussian Naive Bayes: This suggests that the software takes the normal distribution of continuous variables for granted.

2. Multi-nomial Naive Bayes: Suitable for discrete features, this algorithm can be used to classify texts into several groups.
3. Bernoulli Naive Bayes: Depending on the data, this algorithm is most frequently employed for binary features.

Nevertheless, it yields a solid and frequently highly effective result and serves as a foundation for further strategies. Its strengths include its very short training and prediction times, its ability to handle missing values, and its performance with high-dimensional data. But it is predicated on the idea that features are unrelated to one another, as a result, it is limited in situations where several features are connected in the real world [45].

In application, Naive Bayes will be combined with techniques like feature selection or smoothing. It is typically combined with the TF-IDF vectors or the bag-of-words for text categorization.

3.1.2 Support Vector Machine

Support Vector Machines are considered strong tools of supervised learning applicable to classification and regression methods. The objective of the SVM is to determine the best hyperplane that separates different classes in the space of higher dimensions [46].

The fundamental concept of SVM is to choose a hyperplane that has the maximum margin between the classes. For a binary classification problem, the decision boundary can be represented as:

$$\mathbf{w} \cdot \mathbf{x} - b = 0, \quad (3.3)$$

where x is the input vector, w is the normal vector to the hyperplane and b is the bias. The classification rule is then:

$$f(x) = \text{sign}(w \cdot x - b). \quad (3.4)$$

The SVMs use the concept of the support vectors which are, the data points which are closest to the decision line. The optimisation problem for finding the maximum margin hyperplane can be formulated as

$$\text{minimise} \quad \frac{1}{2} \|w\|^2 \quad (3.5)$$

subject to

$$y_i(w \cdot x_i - b) \geq 1 \quad \text{for all } i, \quad (3.6)$$

where y_i are the class labels (-1 or 1) and x_i are the input vectors.

In the case of non-linearly separable data, SVMs use what is known as the "kernel trick" wherein the data is transformed from the input space into a higher-dimensional feature space, where the classes are separable through a linear function [47]. Common kernel functions include

- Linear: $K(x, y) = x \cdot y$,
- Polynomial: $K(x, y) = (\gamma x \cdot y + r)^d$,
- Radial Basis Function (RBF): $K(x, y) = \exp(-\gamma \|x - y\|^2)$.

SVMs can be used in multi-class classification by using methods like one-vs-rest or one-vs-one. They can also be adapted for regression tasks (SVR) where it also incorporates an ϵ -insensitive loss function.

The benefits of SVMs include the ability to work with high-dimensional space and occupy less memory space and a variety of kernels. However, depending on the type of features used, they can provide less sensitivity to some aspects such as feature scaling and may often need good parameter tuning [48].

3.1.3 Random Forest

Random Forest is another kind of ensemble learning technique in which big numbers of decision trees are created and then results are aggregated. Breiman presented it in 2001, and since then, it has been considered one of the most efficient and widely used approaches to machine learning in terms of classification and regression [49].

The primary concept of Random Forest is based on the construction of a set of uncorrelated decision trees and their output fusing. Each tree in the forest is built using the following process:

1. Bootstrap Aggregating (Bagging): A random sample of N cases is selected with replacement from the original data.
2. Random Feature Selection: At each node, among all p features, m features (usually $m \approx \sqrt{p}$) are randomly chosen to split the node.

3. Tree Growing: It means that the best split on those m features is used to split the node.

The final prediction for classification is typically made by majority voting, while for regression, it is the average of individual tree predictions:

- Classification: $\hat{y} = \text{mode}(\hat{y}_1, \hat{y}_2, \dots, \hat{y}_n)$,
- Regression: $\hat{y} = \frac{1}{n} \sum_i \hat{y}_i$,

where $\hat{y}_i$ is the prediction of the i -th tree.

Random Forest introduces two key sources of randomness: original sampling of the data points and random selection of features in the construction of individual nodes. This randomness aids in decorrelating of the trees thereby avoiding overfitting and increasing the model's ability to generalise [50].

Important hyperparameters in Random Forest include:

- Number of trees (n_estimators)
- Maximum depth of trees (max_depth)
- Minimum number of samples required to split a node (min_samples_split)
- Number of features to consider for best split (max_features)

Some of the benefits of Random Forest may include; High accuracy, dealing with large datasets with high dimensions, feature importance estimation and it is very strong with outliers and nonlinearity magnitude. It also offers an internal assessment of the level of generalized error known as Out-of-Bag (OOB) error [51].

The disadvantage of Random Forest is that when the model is built, it is time-consuming and may become 'over-fitted' on some problems as well as noisy data. However, the interpretability of the model is relatively lower than the single decision trees.

3.1.4 XGBoost

XGBoost is the optimised distributed gradient boosting library for 'Scalable Machine Learning'. It was introduced by Chen and Guestrin [52] in 2016 and it slowly gained

the status of the most used machine learning algorithm, especially in the case of structured/tabular data.

As an implement of gradient boosting decision trees (GBDT), XGBoost brings several enhancements for boosting decision trees to better performance and less computation time. The basic concept of building the current round is to combine weak learners - independently, in a sequential manner where every new model attempts to minimise the mistakes of the previous model.

The objective function that XGBoost optimises can be written as:

$$\text{Obj}(\theta) = L(\theta) + \Omega(\theta), \quad (3.7)$$

where $L(\theta)$ is the training loss function and $\Omega(\theta)$ is a total derivation of the model with respect to the parameter θ .

For a given dataset with n examples and m features, the k th tree $f_k(x)$ yields the prediction $\hat{y}_i$ for the i th instance:

$$\hat{y}_i = \sum_k f_k(x_i), f_k \in F, \quad (3.8)$$

where F is the space of regression trees. XGBoost uses a second-order Taylor expansion of the loss function for faster optimisation:

$$L(\theta) \approx \sum_i \left[l_i + g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) \right], \quad (3.9)$$

where g_i and h_i are first and second-order gradient stats on the loss function respectively. Key features of XGBoost include:

1. Regularization: L^1 and L^2 norm to correct for overfitting is used.
2. Handling sparse data: The means used to deal with missing values and zero values.
3. Weighted quantile sketch: For approximate tree learning.
4. Block structure: For parallel learning.
5. Cache awareness: For out-of-core computation.

Some of the benefits of XGBoost are improved prediction rate, suitable for different types of data and loss functions and is computationally fast. It also has the cross-validation and feature importance evaluation integrated [53].

However, an overfitting tendency may be observed with XGBoost if not properly tuned, and its interpretability is a weakness because of the model's intricacy. It may also need more memory than simpler algorithms, should the problem define a large state space [54].

3.2 Deep Learning Approach

The flowchart shown in Figure 3.2 outlines the steps used to carry out fake news detection using Deep Learning models.

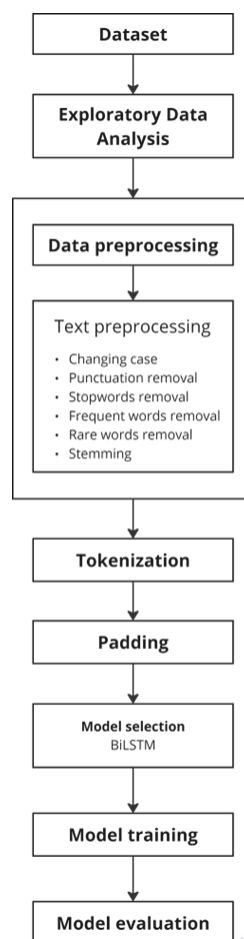


Figure 3.2: Deep Learning methodology.

3.2.1 BiLSTM

Bidirectional Long Short-Term Memory (BiLSTM) is a type of recurrent neural network (RNN) architecture that combines two LSTMs: one that processes the input sequence

in the forward direction and another that processes it in the backward direction. This enables the network to maintain the history of the inputs for each element in the input sequence in terms of both past and future context which makes it useful in tasks such as natural language processing and speech recognition [55].

A key part of a BiLSTM is the LSTM cell, the manipulation of which can solve the vanishing gradient issue in standard RNNs. An LSTM cell comprises three gates: an input gate, a forget gate, an output gate; and a memory cell. The key equations for an LSTM cell are:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f), \quad (3.10)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i), \quad (3.11)$$

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c), \quad (3.12)$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t, \quad (3.13)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o), \quad (3.14)$$

$$h_t = o_t * \tanh(C_t), \quad (3.15)$$

where:

f_t, i_t, o_t are the forget, input, and output gates,

C_t is the cell state,

h_t is the hidden state,

σ is the sigmoid function,

W is the weight matrices and b is the bias vectors.

The architecture of BiLSTM is shown in Figure 3.3. In a BiLSTM, there are two LSTM layers, and while one processes the sequence forward, the other processes it backwards. The forward LSTM ($\rightarrow$) takes the sequence from left to right, and, on the other hand, the backward LSTM ($\leftarrow$) operates from right to left. The outputs of both LSTMs are typically concatenated to form the final output:

$$h_t = [\vec{h}_t; \overleftarrow{h}_t] \quad (3.16)$$

Due to the bidirectional architecture, BiLSTMs are effective in modelling long-distance dependencies from both past and future that is, they find their application in named entity recognition, part-of-speech tagging and sentiment analysis [56].

Some advantages of BiLSTMs include the following: they are capable of utilising bidirectional context, they are good in processing of sequences that vary in length,

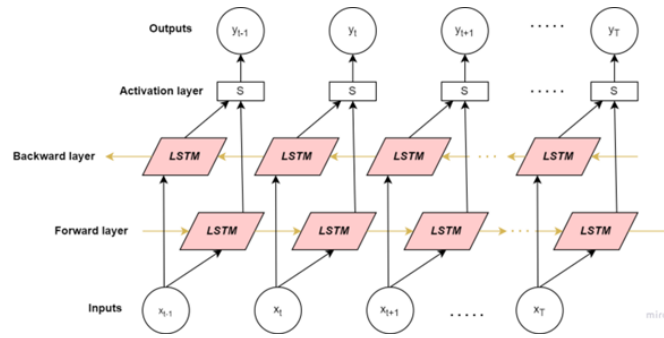


Figure 3.3: Bidirectional Long Short-Term Memory (BiLSTM) architecture.

and they can easily solve long-term dependency problems. However, they can be of high computational cost, demand a large quantity of training data and if not properly regularized can cause overfitting [57].

3.3 Data collection

The "ISOT FAKE NEWS DATASET" utilized in this study was obtained from [Kaggle](#), an open-source data website that is accessible to the public online. The real news stories were gathered from the Reuters.com website and accurately portray global affairs. The fabricated articles were collected from phoney websites such as Wikipedia and Politifact.com, an American fact-checking group.

This dataset comprises 44,898 news articles and was split into two files namely "fake" and "true" files. They were accessible in CSV format. Further, they were compressed within a zip file as "fake.csv" and "true.csv". Each file is comprised of four features namely: title, text, subject and date. The description for each feature is listed below in the Table 3.1. The further analysis of these features is described in the Exploratory Data Analysis Section.

There were certain factors taken into consideration during the selection of an appropriate dataset. To avoid building a dataset with an imbalance of either real news stories or fake news, the dataset built required a balance between the two. Also, a host of news articles that may relate to virtually any topic ranging from international news, and political news, to social media news, among others, were incorporated into the analysis. In addition, the data needed to be accurate and replicate intricate real-world situations to help decipher the patterns involved in spotting false information. Ultimately, a

Feature	Description
title	title of the news article
text	brief description of the news article
subject	types of news (world news, political news, left news, government news, US news, middle-east news)
date	date of the news articles published

Table 3.1: Description of the features in the news article dataset.

reasonable dataset size is necessary; too many records lead to excessive computing demands; and too few records make it harder for the model to discover the underlying patterns. Since only a small number of alternative datasets met each of these criteria, the ISOT dataset was selected for our research.

3.4 Performance Metrics

Performance metrics are extremely important when it comes to evaluating and comparing different ML models. They are qualitative measures, which specifically give out quantities of the ability of the model in its intended task. The metric to be used would depend on the particular problem at hand and the relative importance of different types of errors. In classification problems, some of the measures in use are Accuracy, Precision, Recall, F1-score and the confusion matrix.

1. **Accuracy:** Accuracy means the total number of correctly predicted cases including true positive and true negative divided by total number of cases that have been analyzed. Despite this logic, accurate results can be quite deceptive especially when working with big data with imbalanced distribution.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (3.17)$$

where:

TP = True Positives,

TN = True Negatives,

FP = False Positives,

FN = False Negatives.

2. **Precision:** Precision refers to the measure of instances that are accurately or correctly classified as positive. It answers the question: ‘Among all the instances the model has classified as positive, how many are positive?’ In layman’s terms, high precision equals to low percentage of False Positives.

$$\text{Precision} = \frac{TP}{TP + FP}. \quad (3.18)$$

3. **Recall:** Recall, referred to as sensitivity, or true positive rate, in this context is the capacity of the model to identify all the positive samples. It answers the question: Among all the actual positive instances available, how many were correctly captured by the model? High recall is essential where a ‘miss’ on a positive can be highly costly.

$$\text{Recall} = \frac{TP}{TP + FN}. \quad (3.19)$$

4. **F1-Score:** F1-score is a single score that has emerged from the metrics of precision and recall. It is particularly useful when the class is distributed unevenly. F1-score is the harmonic mean of precision and recall, providing a more balanced measure than the arithmetic mean. However, it is particularly beneficial when you need to scale the priorities between high accuracy and recall, and classes are imbalanced.

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (3.20)$$

5. **Confusion Matrix:** The confusion matrix provides a comprehensive view of a model’s performance across all classes. It’s particularly useful for multi-class classification problems. The structure of the confusion matrix is shown in Figure

3.4. From the confusion matrix, you can derive the following metrics:

- **True Positives (TP) or Sensitivity or Recall:** It measures the proportion of actual positive cases that were correctly identified.
- **False Positives (FP):** It measures the proportion of actual negative cases that were incorrectly classified as positive.
- **True Negatives (TN) or Specificity:** It measures the proportion of actual negative cases that were correctly identified as negative.
- **False Negatives (FN):** It measures the proportion of actual positive cases that were incorrectly classified as negative.

		Predicted	
		Positive (0)	Negative (1)
Actual	Positive (0)	TP	FN
	Negative (1)	FP	TN

Figure 3.4: Confusion Matrix. (TP = True Positives, TN = True Negatives, FP = False Positives, FN = False Negatives)

3.5 Technical Requirements

The below setup was made and used, providing a robust environment for conducting this research on fake news detection using ML and DL models.

1. Hardware Requirements:

- A personal computer system with a multi-core processor (Intel evo i5)
- 8GB RAM

2. Software Requirements:

- Operating System: Windows 11

3. Programming Languages and Libraries:

- Python as the primary programming language
- Machine Learning libraries: scikit-learn, XGBoost
- Deep Learning frameworks: TensorFlow 2.x
- Natural Language Processing (NLP) libraries: NLTK
- Data manipulation: pandas, NumPy

- Data visualization: Matplotlib, Seaborn

4. **Cloud Computing:**

- Google Colab, a cloud computing platform for training large models

5. **Documentation and Reporting:**

- LaTeX for writing the dissertation (Overleaf)
- miro.com for creating figures

Analysis and Design

4.1 Exploratory Data Analysis (EDA)

Exploratory Data Analysis plays the role of obtaining preliminary information that will affect further modelling. This section outlines the properties of the dataset and reflects on the findings of calculated patterns and associations of real and fake news articles. Exploratory research involves discovering the patterns and structure in the data by calculating measures and presenting them graphically. EDA facilitates transformations in feature selection and modelling by showing essential characteristics of the data. It also helps to facilitate better comprehension of the data as well as the recognition of relevant challenges and biases towards creating more resourceful fake news detection models[58].

True news dataset: The sample dataset is shown in Table 4.1. The true news dataset consists of four attributes: title, text, subject, and date. It has a total of 21,417 records. All four are of the "object" data type. Among the 21,417 entries in the title column, there are 20,826 unique entries, which suggest 591 duplicates. The text column can also be observed to have a total of 225 duplicated values. The section on dataset preprocessing also describes how these duplicates are dealt with. The subject column has two distinct types: From the domain of political news, 11,272 articles were retrieved whereas 10,145 articles were obtained from the domain of world news. This is represented in Figure 4.1.

	Title	Text	Subject	Date
0	As U.S. budget fight looms, Republicans flip t...	WASHINGTON (Reuters) - The head of a conservat...	Politics	31/12/17
1	U.S. military to accept transgender recruits o...	WASHINGTON (Reuters) - Transgender people wil...	Politics	29/12/17
2	Senior U.S. Republican senator: 'Let Mr. Muell...	WASHINGTON (Reuters) - The special counsel in...	Politics	31/12/17
3	FBI Russia probe helped by Australian diplom...	WASHINGTON (Reuters) - Trump campaign advise...	Politics	30/12/17
4	Trump wants Postal Service to charge 'much mor...	SEATTLE/WASHINGTON (Reuters) - President...	Politics	29/12/17

Table 4.1: Sample of True news data.

Fake news dataset: The sample dataset is shown in Table 4.2.

	Title	Text	Subject	Date
0	Donald Trump Sends Out Embarrassing New Year...	Donald Trump just couldn't wish all Amer...	News	31/12/17
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Devin N...	News	31/12/17
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwau...	News	30/12/17
3	Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announced that...	News	29/12/17
4	Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day me...	News	25/12/17

Table 4.2: Sample of Fake news data.

The fake news dataset contains 23,481 entries of news articles and consists of four features: title, text, subject and date. The type of all four attributes is an "object" which means that the data of the attributes can contain values of any type - string, integer, and

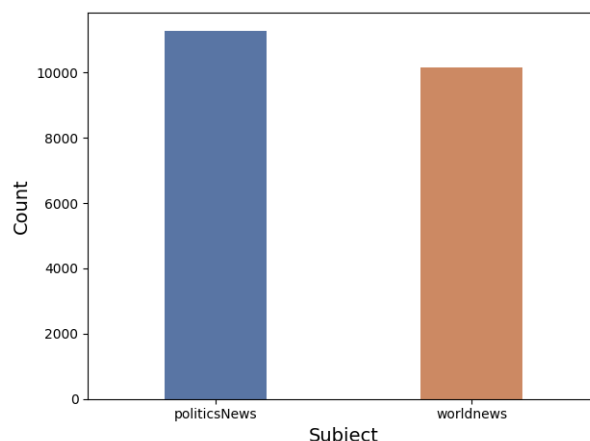


Figure 4.1: Distribution of the subjects in True news dataset. (blue - politics news, orange - world news)

others. The title column itself has a count of 17,903 out of the total records of 23,481. This explains the 5,578 records that have been duplicated. Likewise, the text column is found to have a count of 6,026 duplicate values.

The Subject column has six distinct forms. Based on the nature of the articles, the category called "News" has the most numbers, which is 9,050 articles. The topic of "Politics" is a very vast subject. It is widely covered with a total number of articles equal to 6,841 and it can be also stated that this journal is oriented on subjects connected with political affairs. The subject "left-news" contains 4459 articles total, confirming the signs of the existence of a separate position within the news environment. The category, which is marked as "Government News", contains articles amounting to 1,570, whereas the "US_News" and "Middle-East" contain 783 and 778 articles respectively. Figure 4.2 illustrates the distribution of the above categories. The date column has amorphous formatting, thus complicating the conduct of inference. Hence, all the irregularities for which there exists a groundbreaking solution are discussed as well as visualised in the data preprocessing stage.

4.2 Data Preprocessing

Data preprocessing is a critical step in the data analysis process or data analysis life cycle. It alludes to strategies of cleaning the initial data collected and reorganizing the data so that they can be assimilated into the machine learning frameworks and analysis

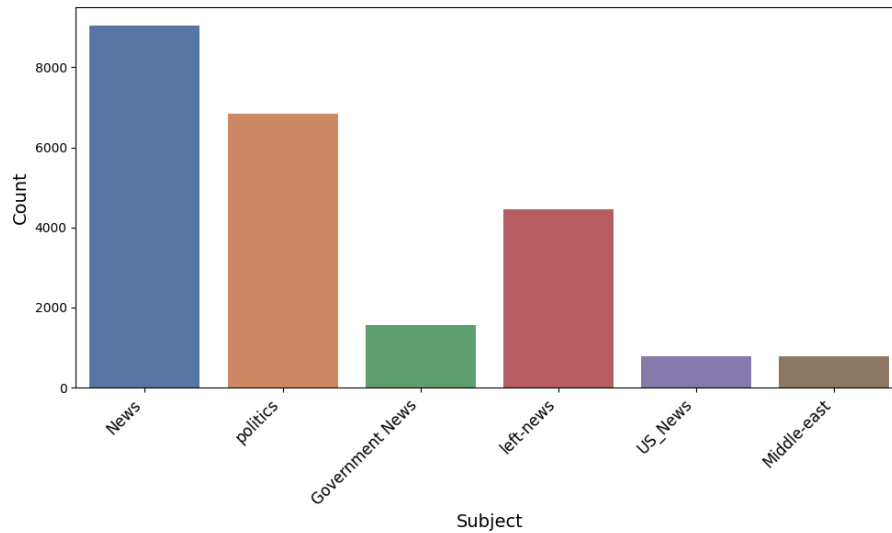


Figure 4.2: Distribution of the subjects in Fake news dataset. (blue - News, orange - politics, green - Government News, red - left-news, purple - US_News, brown - Middle-east)

[59]. Preprocessing is an important step in data analysis because proper handling of the data will help to improve the quality of the derived insight and boost the performance of predictive models.

Labelling the datasets: Data in machine learning is assumed to be pre-tagged in such a way that the machine learning algorithms can learn from and then produce the required predictions. Some of the data existing in the study does not include a feature that defines whether the news is fake or real. As a result, it can be drawn to a close that CSV files contain only true news articles and fake news articles distinctly. Thus, a new column is included in each of the datasets and labelled as the classification label. Since fake news is the classification to be predicted, it is assigned the class value 1, and true news is assigned 0. Such labelling enables categorisation and, therefore, identification of different types of data in a particular dataset. It is illustrated in Tables 4.3 and 4.4.

Joining the datasets: The true and fake data frames are concatenated based on their index and stored in another dataframe. This measure is applied to achieve the objective of compiling data that originates from more than one file in order to co-relate the data and create a meaningful output. However, it offers a detailed framework to embark on additional research on the same topic. Then the dataframe is randomly rearranged to reduce biases and over-train the machine learning algorithm that was developed.

Removing duplicate values: In the combined dataset, there are 6169 duplicates in

	Title	Text	Subject	Date	Label
0	Donald Trump Sends Out Embarrassing New Year...	Donald Trump just couldn't wish all Amer...	News	31/12/17	1
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Devin N...	News	31/12/17	1
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwau...	News	30/12/17	1
3	Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announced that...	News	29/12/17	1
4	Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day me...	News	25/12/17	1

Table 4.3: Label = 1 for Fake dataset.

	Title	Text	Subject	Date	Label
0	As U.S. budget fight looms, Republicans flip t...	WASHINGTON (Reuters) - The head of a conservat...	politics	31/12/17	0
1	U.S. military to accept transgender recruits o...	WASHINGTON (Reuters) - Transgender people wil...	politics	29/12/17	0
2	Senior U.S. Republican senator: 'Let Mr. Muell...	WASHINGTON (Reuters) - The special counsel in...	politics	31/12/17	0
3	FBI Russia probe helped by Australian diplom...	WASHINGTON (Reuters) - Trump campaign advise...	politics	30/12/17	0
4	Trump wants Postal Service to charge 'much mor...	SEATTLE/WASHINGTON (Reuters) - President...	politics	29/12/17	0

Table 4.4: Label = 0 for True dataset.

the title column and 6252 in the text column. To begin with the first step of cleaning since the subsequent extermination focuses on different columns of the duplicates, the text column is cleaned, focusing on which many features will be extracted in the future. This leads to the operation where a few rows containing comparative values in the title column will also be deleted. However, it is noticed that still there are some records

where the title has duplicate values present in it.

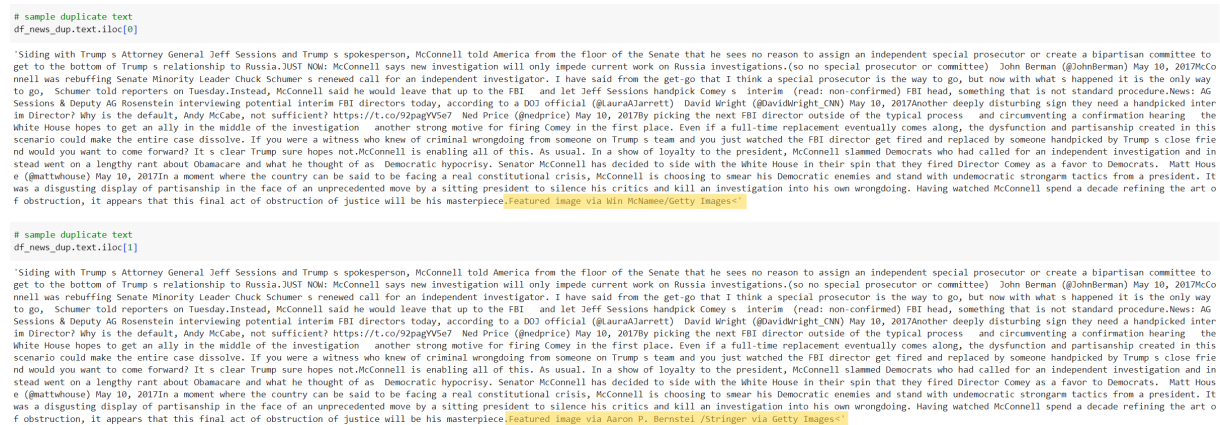


Figure 4.3: Snapshots of the duplicate texts from the Python interface.

	Title	Text
Before removing duplicates	6169	6252
After removing text duplicates	376	0
After removing title duplicates	0	0

Table 4.5: Removing duplicates.

There are slight differences with regard to the description of news as depicted in Figure 4.3, which includes punctuation, news source, image source, etc. To begin with, one gets the impression that the title values are the same as the values in the text column. However, when paying more attention, you realize that such values are identical. Thus, the removal of duplicate titles has led to the removal of duplicate texts. The count of duplicates is stated in Table 4.5.

Date formatting: The dates in the dataset were discovered to be in a variety of forms during analysis, including "January 29, 2017" and "Aug 10, 2016." Furthermore, formatting issues resulted in sentences and hyperlinks in the dataset. Consequently, words and connections were removed initially, leaving only the date numbers. Afterwards, the dates were converted to "Month, Day, Year" format (example: January 15, 2023). Figure 4.4 shows the distribution by the year the article was published. The graph indicates that 36.2% of the news articles were published in 2016, while the remaining 59.5% articles were published in 2017.

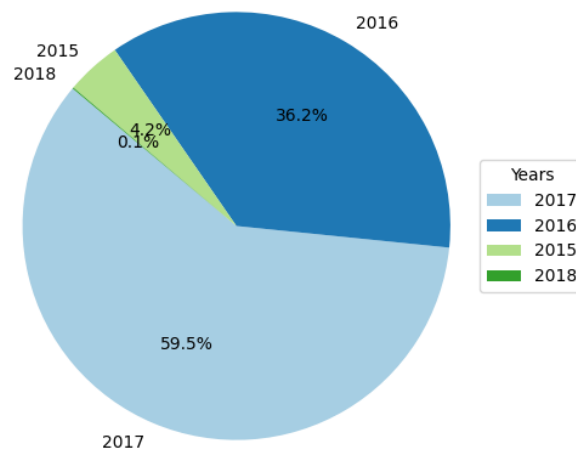


Figure 4.4: Year of publication of news articles.

Renaming and encoding the subject column values: After merging the two files, it was found that the subject column contained eight different categories, two of which had the same meaning but different names. The category "News" is renamed as "World News" and the category "Politics" as "Political News" to maintain consistency with the subject column's values. The dataset currently includes the following subject categories: "world news", "politics news", "left news", "Government news", "Middle-east news", and "US news". The standardised dataset produced by renaming the subject categories is appropriate for carrying out additional analysis. Table 4.6 presents the 6 subject categories and their value counts that we have after renaming.

Subject	Count
world news	18,868
politics news	15,326
left-news	2,394
Government News	893
Middle-east	415
US_News	368

Table 4.6: Subject Value Counts.

Now the subject column contains six distinct categorical values which cannot be directly fed to train or predict the model as it is in textual form. So, now feature encoding

is done to make it acceptable for the model to work. The use of the "get_dummies()" function creates new columns regarding categories found in the subject column. These columns consist of binary values (0 for false and 1 for true) which show whether a particular data instance belongs to a particular category. This transformation is useful when converting categorical values into numerical forms, making them suitable for use as features in the training process.

4.3 Text Preprocessing

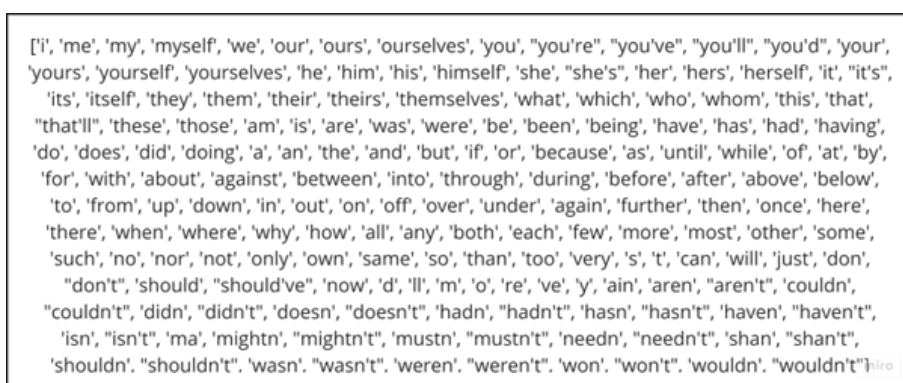
In text analysis, text data must be preprocessed before they can be used in text or natural language processing (NLP) tasks. It involves transforming text-based data that is usually messy into one with better structure and analysis [62]. Tokenisation, stemming or lemmatisation, converting to lowercase, identifying and eliminating special characters and punctuations, eradicating stop words, and handling contractions are examples of preprocessing steps [61]. Hence, the text data can be preprocessed to remove noise, formatted in a more standard way and prepared for further usage or for feeding in a machine-learning model. The following text preprocessing steps were applied to the text data.

Converting text to lowercase: Converting the text into lowercase is one of the simplest and fundamental preprocessing techniques in natural language processing. This process includes converting all uppercase letters in a text into their equivalent lowercase letters. The purpose is to standardise the text, thereby minimising its variation, and making different cases of the word have the same form (e.g., "News", "news", and "NEWS") [62].

Punctuation removal: The process of removing punctuation is one of the most significant steps in text preprocessing. This function makes use of regular expressions that help to eliminate different factors that may systematically hinder text analysis. It begins with stripping the URLs and HTML tags, and further filters text between square brackets. The function removes all non-word characters and replaces them with spaces and also discards all punctuation marks. It also deletes newline characters and words with numbers. It is a more detailed way of cleaning the data, resulting in plain text that is more consistent and therefore easier to process in further natural language processing

tasks.

Stop words removal: The removal of stop words is a crucial text preprocessing method where unimportant words that occur frequently in a text are excluded. These words include "the", "is", "and", "in" among others and in most cases they appear regularly in different texts but they do not have much impact on the actual content or even the tone of the message being passed. A list of stop words is illustrated in Figure 4.5.



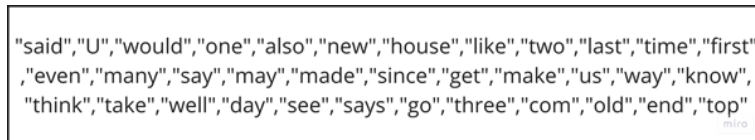
[i', 'me', 'my', 'myself', 'we', 'our', 'ours', 'ourselves', 'you', 'you're', 'you've', 'you'll', 'you'd', 'your', 'yours', 'yourself', 'yourselves', 'he', 'him', 'his', 'himself', 'she', 'she's', 'her', 'hers', 'herself', 'it', 'it's', 'its', 'itself', 'they', 'them', 'their', 'theirs', 'themselves', 'what', 'which', 'who', 'whom', 'this', 'that', 'that'll', 'these', 'those', 'am', 'is', 'are', 'was', 'were', 'be', 'been', 'being', 'have', 'has', 'had', 'having', 'do', 'does', 'did', 'doing', 'a', 'an', 'the', 'and', 'but', 'if', 'or', 'because', 'as', 'until', 'while', 'of', 'at', 'by', 'for', 'with', 'about', 'against', 'between', 'into', 'through', 'during', 'before', 'after', 'above', 'below', 'to', 'from', 'up', 'down', 'in', 'out', 'on', 'off', 'over', 'under', 'again', 'further', 'then', 'once', 'here', 'there', 'when', 'where', 'why', 'how', 'all', 'any', 'both', 'each', 'few', 'more', 'most', 'other', 'some', 'such', 'no', 'nor', 'not', 'only', 'own', 'same', 'so', 'than', 'too', 'very', 's', 't', 'can', 'will', 'just', 'don', 'don't', 'should', 'should've', 'now', 'd', 'll', 'm', 'o', 're', 've', 'y', 'ain', 'aren', 'aren't', 'couldn', 'couldn't', 'didn', 'didn't', 'doesn', 'doesn't', 'hadn', 'hadn't', 'hasn', 'hasn't', 'haven', 'haven't', 'isn', 'isn't', 'ma', 'mightn', 'mightn't', 'mustn', 'mustn't', 'needn', 'needn't', 'shan', 'shan't', 'shouldn', 'shouldn't', 'wasn', 'wasn't', 'weren', 'weren't', 'won', 'won't', 'wouldn', 'wouldn't']

Figure 4.5: List of stopwords. The picture is generated on Miro.

The elimination of stop words aids in minimizing noise in the data; the dimensionality of the text corpus is also reduced; emphasis shifts towards richer content carrying words. This process usually entails the matching of each word within the text with a stop word list and excluding any matches. The resulting text is shorter and sometimes increases performance.

Frequent words removal: The frequent word removal in text preprocessing begins by using a Counter object to tally word occurrences across all texts in the dataset. The 200 most common words are then identified. From this list, a subset of frequently occurring words is manually selected for removal. This subset of words is illustrated in Figure 4.6. These words are not stopwords but are also excluded as they seem to be of low value for the particular analysis. The `discard_frequent_words()` function then excludes these selected frequent words from the texts. It aids in narrowing down a large number of highly frequent and less meaningful words in the context of the topic to be analyzed and focus on more relevant terms that can provide more valuable information about the topic.

Rare words removal: Eliminating low-frequency terms could help the reader to



"said", "U", "would", "one", "also", "new", "house", "like", "two", "last", "time", "first",
"even", "many", "say", "may", "made", "since", "get", "make", "us", "way", "know",
"think", "take", "well", "day", "see", "says", "go", "three", "com", "old", "end", "top"

Figure 4.6: Manually selected list of frequently occurring words. The picture is generated on Miro.

pay more attention to more frequent terms that may be more significant in some way and could be useful in terms of saving time and optimising the impact. Preprocessing text data by elimination of rare words begins with the process of making a set of the rare words list which includes any word that has occurred not more than 150 times in the whole corpus when computed as per the word count. The `discard_rare()` function eliminates these rare words from each text. This process seeks to minimise noise and dimensionality within the data through the exclusion of words that rarely feature and therefore may not enhance textual analysis significantly.

Stemming: Stemming is a preprocessing technique in natural language processing whereby words are reduced to their base or root forms[63]. The `stemming_text()` function breaks the input text into separate words and then applies the stemmer on each of the words. This involves the removal of suffixes and, at times, the prefix in order to arrive at a base form of a word. Stemming helps in the reduction of the vocabulary because words with similar meanings are processed as a single term and variations of a word are treated as the same.

4.4 Data Visualization

Word clouds in this aspect display the most utilized words in various sets of news data, possibly concerning the political events of Americans and their elections. Figure 4.7 contains information from both true and fake news sources. Figure 4.8 illustrates word cloud for true news while Figure 4.9 illustrates a word cloud for fake news.

In all three, some keywords such as ‘president’, ‘trump’, ‘state’, ‘clinton’ appear many times, pointing out that these items were among the most widely discussed topics in the news. The differences thus obtained from word clouds can help in understanding how language and focus shift between true and fake news sources. For example, the frequency of the words may differ, which may suggest shifts in the position or

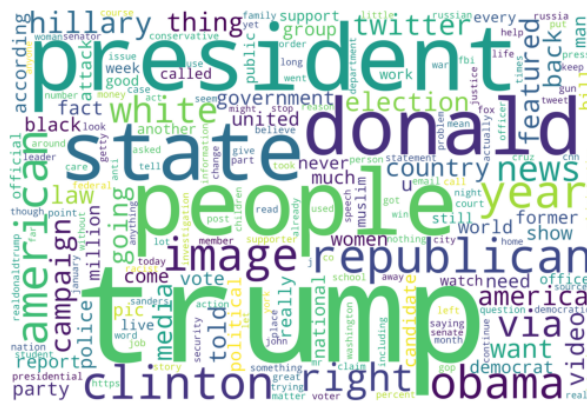


Figure 4.9: Word cloud for fake news dataset.

technique is based on the use of prior knowledge and creativity to obtain additional enriched representation from raw data. Transformation can include a process of feature creation where for example one can generate new features from the already existing ones by doing things like combining features, applying logarithmic or polynomial of a variable or even encoding categorical variables, or creating interaction terms [65]. The purpose is to extract features of the data that may not be obvious, allowing an easier task for learning employing machine learning algorithms. Feature engineering can have a major positive impact on model performance, mitigate the risk of overfitting and offer a better understanding of the patterns inside the data [66]. Most often, it is viewed as a very subjective task that combines a set of well-developed algorithms and intuition about the problem domain.

4.5.1 Feature Extraction

Feature extraction is the process of selecting and transforming the raw data into the features that are relevant for model training and/or further data analysis [67]. From textual data, the following features are extracted for model training and evaluation.

Textual Features

- **Uppercase count:** This feature measures the number of capital letters present in a text. In sensational or fake news articles the headlines are usually written in all capital letters to attract the attention of the reader. Thus, a higher count of uppercase letters might suggest the ‘bias of emotions’ or an effort to portray something as more important than it is, which is a characteristic of fake news. However, it

should be mentioned that other news sites may also employ capitals for acronyms or to signify emphasis, which makes this criterion relevant in conjunction with others [1].

- **Lowercase count:** This feature involves the total count of all the lowercase letters in a given text. If the ratio of the percentage of lowercase characters to that of the overall percentage of characters is higher, it could suggest that the text may be less formal and more likely to be typical of user-created or news articles from suspicious sources.
- **Punctuation count:** This feature measures the numbers and type of punctuation marks that are used in a text. It can help extract information about writing style, choice of words, and possibly even the reliability of the news articles. The repetition of certain punctuations like exclamation points or question marks can have an impact on the readers as such is considered sensationalism which contributes to the construction of fake news[68]. On the other hand, moderate use of these punctuations could represent formally professional writing as is characteristic of many genuine news outlets. In the articles identified as fake news, there are more punctuation marks especially exclamation marks than those in articles identified as real news articles [69].
- **Character count:** This feature counts the number of characters that are contained in a text and these include letters, numbers, symbols, as well as spaces. In the case of fake news detection, this feature may help to understand the length and relative complexity of articles. Fake news is typically shorter than real news since creating long content consumes more time and effort than the creation of short material. On the other hand, very long articles might signify a detailed study common to credible news sources [70].
- **Word count:** This feature indicates the total number of words in a text and informs about the length and possibly complexity of the article. Word count is also a useful measure in the case of fake news detection. Fake news content is shorter than real news stories on average. It might be because of the time and energy needed to create elaborate, extensive content or the goal of briefly grabbing the reader's attention with specific, emotion-evoking information [70].

- **Sentence count:** This feature provides the number of sentences in a text, which can be useful in understanding the distribution and density of an article. In the context of fake news detection, the use of this feature can be associated with the writer's style as well as the content depth. Compared to legitimate news, fake news includes fewer sentences, perhaps because they are crafted in a more concise format that is designed to grab the attention [70].
- **Syllable count:** This feature highlights the total number of syllables in the textual content and is useful to determine the text's complexity and ease of readability. In the case of fake news detection, syllable count can be an influential indicator of linguistic complexity. Fake news is written in less complex language to capture the attention of a large number of people and elicit a response from these people. Generally, less than 1 syllable per word could be an indicator of lower readability, or oversimplification of the content which is a typical feature of fake news. A higher syllable count could imply more complex language which is seen as appropriate for professional and credible journalism, which is typically associated with research-based articles [71].
- **Flesch reading ease:** This feature is a readability metric that evaluates how simple it is for a reader to understand a given text. In fake news detection, this feature can provide insights into the intricacy and readability of news articles. The language used in fake news is simpler compared to that in real news. Flesch Reading Ease = 0 to 100, with 0 being difficult and 100 being easy, determined by the average number of words per sentence and syllable per word [70].
- **Part-of-Speech (POS) tagging:** This feature is an NLP procedure that aims to assign a grammatical analysis label (noun, verb, adjective and adverb etc) to every word in a text. In the context of fake news detection, POS tagging can capture linguistic patterns of the written language that may significantly set samples from fake news and real ones apart from one another. The distribution of POS tags can differ in truthful and deceptive content, where fake news often uses pronouns, adverbs, and verbs, while legitimate news includes more often nouns [21]. A powerful NLP library called spaCy is used to perform POS tagging on each text. It counts the occurrences of each POS tag and then creates separate columns for

each tag type in the DataFrame as shown in Table 4.7.

ADJ	ADP	ADVAUX	CCONJ	DET	INTJ	NOUN	NUM	PART	PRON	PROPN	SCONJ	VERB	
19.0	5.0	1.0	0.0	1.0	0.0	0.0	73.0	0.0	0.0	0.0	64.0	0.0	21.0
8.0	0.0	1.0	1.0	2.0	0.0	0.0	36.0	0.0	0.0	0.0	48.0	0.0	18.0
5.0	2.0	1.0	0.0	0.0	0.0	1.0	26.0	0.0	0.0	0.0	37.0	0.0	6.0
25.0	2.0	4.0	3.0	0.0	0.0	0.0	124.0	2.0	0.0	0.0	95.0	1.0	30.0
21.0	0.0	1.0	2.0	0.0	0.0	1.0	83.0	0.0	0.0	0.0	155.0	1.0	24.0

Table 4.7: Part-of-Speech Tagging.

- **Word2Vec:** Word2Vec is one of the most efficient word embedding processes that aims to map words into dense vectors in a continuous vector space in order to show the semantic relation between them. Word2Vec can be useful in fake news detection since while performing this task, it helps capture the contextual meaning of the words and phrases within the news articles. Word2Vec was utilised to train with a vector size equal to 100, and a window of 2 on the tokenized texts. It then creates document-level embeddings by averaging the word vectors for each text. The resulting document-level embeddings are used as features in the detection models.

Sentiment Features measure a text's subjective content and emotional tone. In fake news detection, these features can be very important as they complement other measures by highlighting the effect that is contained in some fake news that may not be in real news. There is a higher sentiment polarity with fake news as opposed to genuine news [72]. The following sentiment features were extracted from text data.

- **Subjectivity and Polarity:** Subjectivity and polarity are two sentiment features that have been largely used when classifying fake news. Subjectivity quantifies the extent to which text contains opinions, beliefs or emotions as opposed to facts. Polarity, on the other hand, defines the sentiment of a text as positive or negative. Fake news articles have significantly higher subjectivity and more extreme polarity scores than genuine news[1]. TextBlob which is one of the most commonly used libraries in Python, is used for sentiment analysis to get the subjectivity as well as

polarity of each news article. Subjectivity scores lie between the range of 0 and 1; where 0 is the most objective, and 1 is being most subjective, while polarity scores range between -1 and 1 with -1 being the most negative and 1 being the most positive.

- **Positive and Negative Word Percent:** The sentiment features that measure the ratio of the positive or negative verse in the text are the positive and negative word percentages. In fake news detection, these features can serve the important purpose of giving a general idea about the valence of the article. Fake news tends to include more percentage of the words that can elicit an emotional response among its readers[73]. To arrive at these percentage values, the text data is tokenised, analysis of the polarity of each word is done with the help of TextBlob, and then calculated the percentage of words with polarity greater than zero (positive) or less than zero (negative), respectively. A higher percentage of positive or negative words might be influenced by sensationalism or emotion, which are both often associated with fake news.

4.5.2 Feature Scaling

Feature scaling re-scales the data of the independent variables or features so their maximum value is 1. In this regard, MinMaxScaler is used where the features are scaled between a fixed range between 0 and 1. The MinMaxScaler uses the following formula to scale each feature:

$$X_{\text{scaled}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}, \quad (4.1)$$

where:

X is the original feature value,

$X_{\min}$ is the minimum value of that feature,

$X_{\max}$ is the maximum value of that feature,

X_{scaled} is the resulting scaled value.

This is vital as it helps ensure that all features help in learning and that there is no domination of larger magnitude features over the smaller magnitude features. [50] Scaling enhances the convergence of machine learning algorithms and is particularly vital for algorithms that are sensitive to the scale of input features, especially gradient

descent-based algorithms as used in this research.

4.5.3 Feature Selection

Feature selection is the process of choosing the most optimal features to work with, which increases predictive ability, decreases the impact of noise and increases the model interpretability. In this analysis, Mutual Information (MI) is used as the Feature Selection method. Mutual Information quantifies how much one variable depends on the other and can be utilized for estimating how effective a feature is in predicting the target variable[67]. Features with high mutual information scores are most related to the target variable. This approach could also help in dimensionality reduction, and improve model efficiency and accuracy by using only the most important features [74]. Thus, the top 15 features with high MI scores as shown in Table 4.8 along with the Word2Vec feature were selected for further training the machine learning models.

Features	MI Score
Flesch_reading_score	0.103552
char_count	0.072151
syllable_count	0.068596
word_count	0.067048
lc_count	0.061852
subject_left-news	0.055930
PROPN	0.051985
punc_count	0.050102
subject_politicsNews	0.046810
uc_count	0.042894
NOUN	0.037917
VERB	0.036130
ADJ	0.027410
sent_count	0.025465
INTJ	0.023944

Table 4.8: Top 15 features with high Mutual Information (MI) Scores.

4.6 Model Training

A combination of 15 selected features from the mutual information selection with word vector features is used in the training of machine learning models. This creates a comprehensive feature set that includes both content-based and linguistic features. Further, the dataset is split into training (80%) and testing (20%) sets. This split allows for model evaluation on unseen data, helping to assess the model's generalising ability.

In this research, 4 Machine Learning models i.e, Naive Bayes, Support Vector Machine, Random Forest, and XGBoost models and 1 Deep learning model i.e, BiLSTM model trained using the Scikit-learn python library and their performance was evaluated which is discussed in the next chapter.

Naive Bayes: The Naive Bayes model is one of the most common probabilistic classifier models efficiently used in machine learning applications, especially for text classifications such as fake news classification. It is based on Bayes theorem and optimises independence between features which makes the calculations faster and less time-consuming during the training process [75]. In the case of fake news detection, the model is trained on the training data containing the articles and computes its ability to classify the news articles into fake and real news based on the occurrence of some words or features[20]. In practice, Naive Bayes relies on simplistic assumptions but yields good results compared with other strategies, particularly when data volume is big [61].

Support Vector Machine: SVM is one of the most effective supervised learning algorithms used in classification specifically in fake news detection. SVM is based on the principle of identifying the best-fit hyperplane that maximally separates the different classes in the multidimensional feature space [46]. In fake news detection, each news article is a point in this space; features used are often extracted directly from the article text. Hence, the SVM algorithm seeks to identify the hyperplane that creates the largest margin between fake and real news articles. SVM generalizes the solution to the case of non-linearly separable data using kernel functions which map the input space to a higher dimensional feature space. Although the computational complexity of SVM is high, high classification accuracy is possible in text analysis making it important in fake news detection [1].

Random Forest: Random Forest is an ensemble learning method which together with several decision trees are used for creating one model more accurate for classification tasks. When it comes to fake news detection, every tree in the forest is built on a randomly selected portion of the training data and the features [76]. Throughout the training phase, the classification algorithm develops many decision trees which provide a prediction of the class label. The final predicted value is obtained from all the trees by computing the average to minimize overfitting and maximize the model's ability to generalise [51]. Random forest performs quite well when working with high-dimensional data and is able to fit non-linear interactions between the features, which makes it quite useful for classifying texts.

XGBoost: XGBoost stands for Extreme Gradient Boosting which is the advanced implementation of the gradient boosting machine and used for improved efficiency and performance in multiple ML tasks. In regards to fake news classification, the use of XGBoost entails the formulation of an ensemble of weak learners usually decision trees in a sequence manner. Each new tree learns from the mistakes of the previous trees in the ensemble and aims primarily at classifying the boundary samples [1]. It includes a process of adding trees successively over and over again and the result is optimised with the help of a differentiable loss function. Overfitting is avoided in XGBoost by using different techniques including adding the selected regularization terms in the objective function and cross-validation techniques which are inherent in XGBoost [71]. XGBoost can capture non-linear and linear features of textual features in general and specific for fake news detection as well. In addition, feature importance ranking which is a strong point of XGBoost enables it to well perform in the analysis of high-dimensional data. Thus, XGBoost can be seen as especially suitable for pinpointing key indicators of misinformation in news articles [73].

BiLSTM: Let us elaborate on the other preprocessing measures that have been conducted for the training of the deep learning model. In this study, texts were first pre-processed by converting them to the lowercase and then the following operations were performed on the dataset; stopwords removal, removal of rare words, removal of frequent words and stemming. Next, a tokenizer is fit on the training data to create the word-to-index mapping of the terms. Last, the texts of training and testing are transformed into vectors; the values of which are integers that correspond to specific words

of the vocabulary. Next, sequence padding is performed, and thus all the sequences are made to be of equal length, 5000 in this case. This is needed because usually, the inputs to the neural networks must have dimensions of a fixed size. Following that, a Sequential Model is initiated which facilitates the addition of layers sequentially.

First, an embedding Layer is added which transforms integer-encoded words into dense vectors of fixed size (120 dimensions). It assists in the identification of the semantic relevancy of the words. Subsequently, a bidirectional Long Short-Term Memory layer is built in which the sequence is processed for forward and backward contexts to capture information about the past and future.

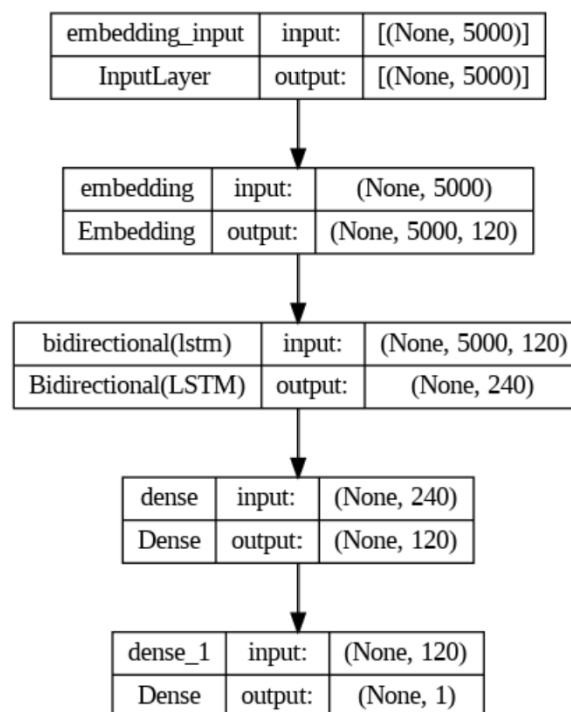


Figure 4.10: BiLSTM model configuration plot.

Next, two Dense Layers are introduced. The first with ReLU activation introduces non-linearity into the model whereas the last layer with sigmoid, outputs a value between 0 and 1. Using this sigmoid function the probability of the occurrences is calculated and 1 is given for fake news and 0 for true news. Figure 4.10 gives a clear picture of all the layers introduced in our sequential model. Last of all, the model is compiled using Adam optimiser, binary cross-entropy, which is used in binary classification tasks, and accuracy as the measure of the model's performance.

4.7 Feature Importance

The need for feature selection was noted earlier in the feature engineering stage as a method to reduce the dimensionality of the data while still being useful in improving the performance of the model. This was a beneficial process within the pre-processing step; nevertheless, the evaluation of how important features are is a crucial part of the analysis. Feature importance provides the measure of the significance of input variables, which is crucial in fake news detection [1]. Feature importance evaluation normally occurs after developing a model by identifying how each feature affects the outcome. Since random forests and XGBoost are tree-based models, they provide an inherent measure of feature importance which is calculated based on the frequency of usage of the feature for creating splits in the trees [52]. The importance scores are standardized and then ranked so that we can determine which factors could have the most significant impact in separating fake from real news. Not only does this analysis increase model interpretability but also assists in feature selection which leads to higher model performance as well as improved computational speed. This concept of feature importance was employed on the best-performing classification model to determine which features were useful in defining fake news and which ones were genuine. For this purpose, the internal feature importance facility of the model was employed.

4.8 Sentiment Analysis

In the conventional fake news detection strategy, the directed attention is on the syntactic and semantic features of the sentences. It goes a step further to study sentiments which have received relatively little attention in research. Four sentimental features were statistically identified and extracted and the best-performing model was tested solely for these features. This is important in establishing the extent to which the model can analyze sentiment features such as the emotional and subjective stance of the news article in identifying fraud news without reference to the other features. Later on, these sentimental features are incorporated in combination with the original features to form a broad feature set and once again the best model is tested on the broad feature set. The study also seeks to discover the extent to which the addition of sentiment will enhance

or hinder the ability of the model to classify fake news from real news using textual language.

Results and Discussions

5.1 Results of Machine Learning Models

- **Naive Bayes:** The Naive Bayes model performance is tabulated in Table 5.1 and the corresponding confusion matrix is shown in Figure 5.1.

Performance Metrics	Score
Accuracy	86.090
Precision	70.194
Recall	98.699
F1-score	82.041

Table 5.1: Performance of Naive Bayes classifier.

The accuracy of the Naive Bayes model is 86.090%, which reveals a high level of effectiveness in identifying fake news. This means that the proposed model can identify fake and real news articles with about 86% efficiency. It is observed that out of all the evaluation metrics, the model achieved the highest recall score of 98% indicating that the model is virtually perfect at identifying fake news articles without missing any. However, the level of precision of 70.194% makes it possible to suggest that only about 70% of the articles that are considered fake are fake. This means that the model tends to over-predict, thereby categorizing some true news as fake news. The F1-score of 82% shows that, generally, the model works

quite well in terms of both precision and recall, though it could be improved to have fewer false positives while keeping an equally high false negative rate for fake news. The confusion matrix gives a clear picture of the classification of fake news predicted by the Naive Bayes classifier. The ROC curve has an AUC of 0.85 indicating that the Naive Bayes model has good discriminative power and performs well above chance in classifying instances, though there is still some room for improvement to reach perfect classification. Figure 5.2 illustrates the ROC curve of the Naive Bayes classifier.

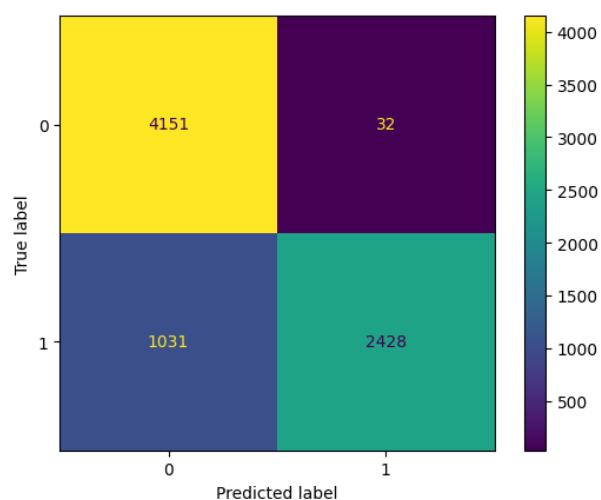


Figure 5.1: Confusion Matrix of Naive Bayes classifier.

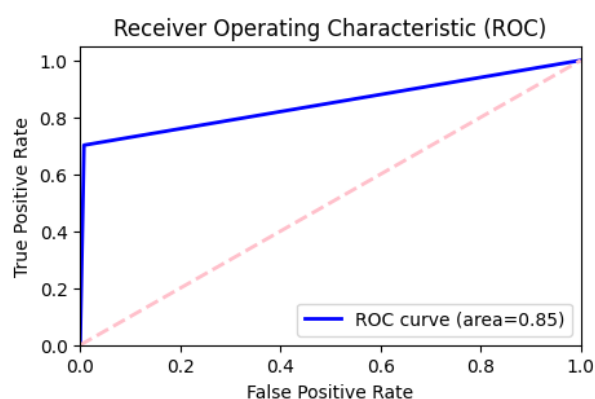


Figure 5.2: ROC curve of Naive Bayes classifier.

- **Support Vector Machine:** The SVM model performance is tabulated in Table 5.2 and the corresponding confusion matrix is shown in Figure 5.3.

Support Vector Machine (SVM) also yields very good results in terms of the ability

Performance Metrics	Score
Accuracy	97.265
Precision	96.328
Recall	97.598
F1-score	96.959

Table 5.2: Performance of SVM classifier.

to identify fake news with an accuracy level as high as 97.265%. This shows that the model gives the right classification of fake and real news articles in instances greater than 97% in both cases. The precision of 96.328% is an affirmation that when the model categorizes an article as fake, it is right in the independent classification 96% the time, implying a low false positive rate. The high recall of 97.598% indicates that the model correctly identifies nearly 98% of fake news articles leaving behind a very small percentage of the overall fake news articles. The F1 score of 96.959%, which takes into account both precision and recall metrics and is an average value, also points to the high performance of the model. These results indicate that the SVM model is highly effective at distinguishing between fake and real news, with a good balance between identifying fake news (recall) and avoiding misclassification of real news as fake (precision). The confusion matrix gives a clear picture of the classification of fake news predicted by the SVM classifier.

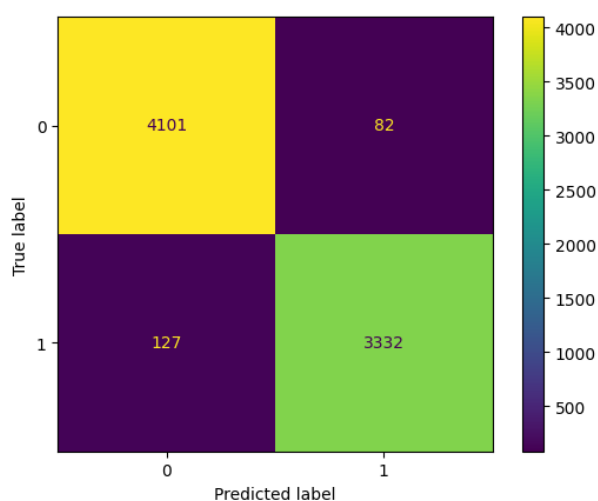


Figure 5.3: Confusion Matrix of SVM classifier.

The ROC curve has an AUC of 0.97 indicating that the SVM model has excellent predictive performance, significantly outperforming random classification (represented by the diagonal dashed line) and approaching optimal classification across various threshold settings. Figure 5.4 illustrates the ROC curve of the SVM classifier.

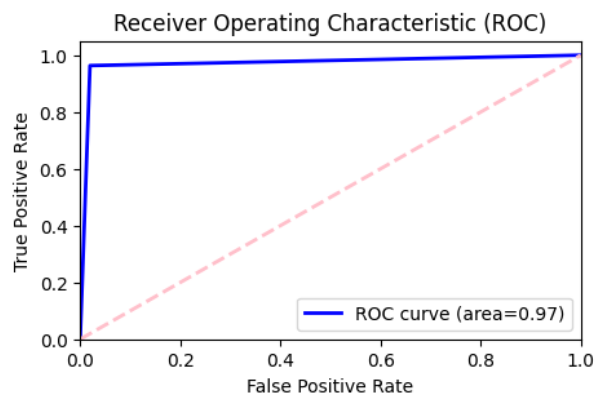


Figure 5.4: ROC curve of the SVM classifier.

- **Random Forest:** The Random Forest model performance is tabulated in Table 5.3 and the corresponding confusion matrix is shown in Figure 5.5.

Performance Metrics	Score
Accuracy	96.100
Precision	95.114
Recall	96.227
F1-score	95.667

Table 5.3: Performance of the Random Forest classifier.

Random Forest model shows great effectiveness in recognizing the fake news with an accuracy of 96.100%. This means that the model can identify fake as well as real news approximately 96% of the time. The precision of 95% indicates that when the model has predicted an article to be fake, it is right in approximately 95% of the cases, implying that the model might have a low percentage of False Positive. The values of recall are as high as 96.227%, which means that the model learns to recognize more than 96% of all fake news articles in the given set, perhaps even missing a few. In addition, the F1-score of 95.667% which takes into consideration

both precision and recall affirms the reliable performance of the model. From this analysis, the Random Forest algorithm demonstrates high accuracy in detecting fake news while simultaneously achieving good levels of recall and precision. Although a little less accurate than the SVM model, these results are still quite good and attest to the Random Forest model being a worthy contender for tasks involving fake news detection. The confusion matrix gives a clear picture of the classification of fake news predicted by the Random Forest classifier.

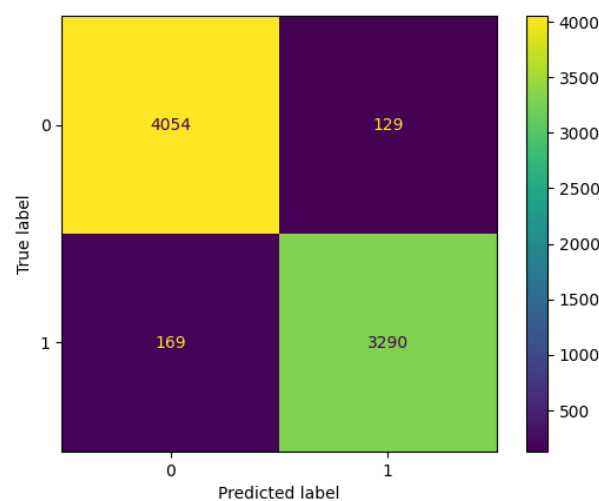


Figure 5.5: Confusion Matrix of the Random Forest classifier.

The high AUC of 0.96 signifies that the Random Forest model significantly outperforms random classification and approaches near-perfect classification. Overall, this ROC curve indicates that the Random Forest model is highly effective and reliable in detecting fake news, with performance almost on par with the SVM model. Figure 5.6 illustrates the ROC curve of the Random Forest classifier.

- **XGBoost:** The XGBoost model performance is tabulated in Table 5.5 and the corresponding confusion matrix is shown in Figure 5.7.

XGBoost model performs best in identifying fake news compared to other machine learning models. It identifies fake as well as the real news data with an accuracy of 98.181% emphasising that it offered very high overall efficiency. The precision of 97.774% implies that if the model classifies an article as fake, it does so with a 98% probability, further implying that the false positive rate is extremely low. The recall of 98.200% also reveals that the model captures more than 98% of all

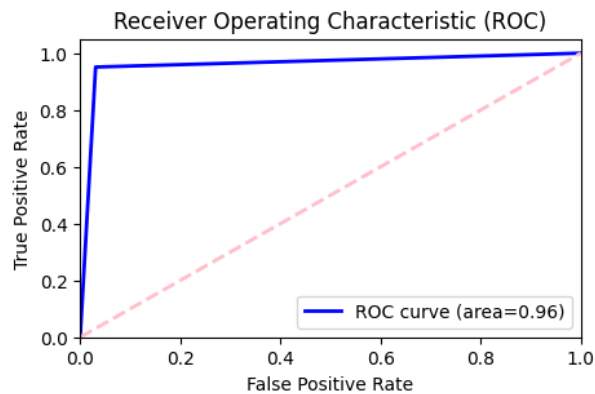


Figure 5.6: ROC curve of the Random Forest classifier.

Performance Metrics	Score
Accuracy	98.181
Precision	97.774
Recall	98.200
F1-score	97.986

Table 5.4: Performance of the XGBoost classifier.

fake news articles in the dataset with minimal omission of fake news. The final metric of F1 score of 97.986% which gives equal importance to Precision and recall supports the overall superior performance of the model. All these results indicate that the proposed XGBoost model is very efficient in the classification of fake and real news with very high measures of recall and precision.

The XGBoost model has a better performance compared to the previous discussion of SVM and Random Forest models and thus is the best among the models examined for this fake news detection task. The confusion matrix gives a clear picture of the classification of fake news predicted by XGBoost classifier. The AUC of 0.98 signifies that the XGBoost model significantly outperforms random classification and comes extremely close to perfect classification. Overall, this ROC curve confirms that the XGBoost model is highly effective and reliable in detecting fake news, showcasing the best performance among all the models analyzed. Figure 5.8 illustrates ROC curve of Random Forest classifier.

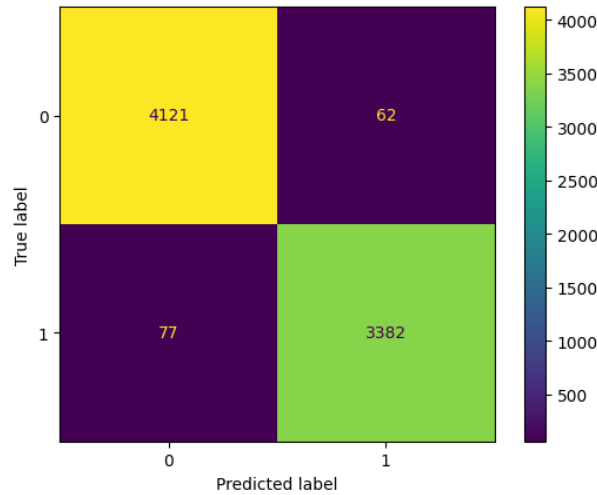


Figure 5.7: Confusion Matrix of XGBoost classifier.

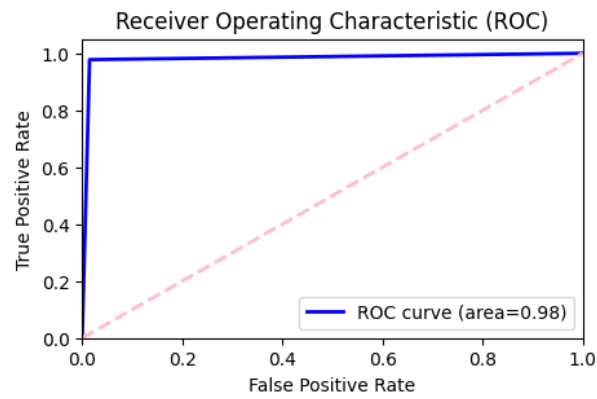


Figure 5.8: ROC curve of XGBoost classifier.

5.1.1 Comparison of the models

Comparing the four models - Naive Bayes, SVM, Random Forest, and XGBoost - for fake news detection reveals a clear progression in performance. The Naive Bayes model, while showing decent results with an accuracy of 86.09% and a high recall of 98.699%, had the lowest overall performance, particularly struggling with precision (70.194%). The SVM model significantly improved upon this, achieving 97.265% accuracy and balanced precision and recall (96.328% and 97.598% respectively). Another remarkable result was achieved with the Random Forest model with slightly lower but still very satisfactory values: accuracy 96.100%, precision 95.114%, recall 96.227%. However, in this experiment, the XGBoost model was ranked as the most effective model as it outperformed all the other models with the highest score of 98.181% accuracy,

	Accuracy	Precision	Recall	F1-score	AUC
Naive Bayes	86.090	70.194	98.699	82.041	0.85
SVM	97.265	96.328	97.598	96.959	0.97
Random Forest	96.100	95.114	96.227	95.667	0.96
XGBoost	98.181	97.774	98.200	97.986	0.98

Table 5.5: Comparison of Machine Learning models.

97.774% precision, and 98.200% recall. This performance hierarchy is also evident in the ROC curve analysis, with AUC scores improving from 0.85 for Naive Bayes to 0.98 for XGBoost. All models with the exception of Naive Bayes proved to work well in the classification of fake news, but the best performance was shown by the XGBoost with slight edges over SVM and Random Forest while Naive Bayes still presented a noteworthy result.

5.2 Results of Deep Learning Model

Bidirectional LSTM model was employed under deep learning to check the effectiveness when compared with machine learning models. Its performance is tabulated in Table 5.6 and the corresponding confusion matrix is shown in Figure 5.9.

Performance Metrics	Score
Accuracy	0.999
Precision	0.999
Recall	1.000
F1-score	0.999

Table 5.6: Performance of BiLSTM classifier.

The performance of the BiLSTM model in detecting fake news is exceptionally high, almost at a level of perfection. It has shown an accuracy of 0.999, the model classified 99% of instances accurately. The F-score that incorporates both precision and recall is also equal to 0.999, which indicates high overall effectiveness. The recall of 1.000 indicates that the model correctly picked all the actual fake news without missing any

of them, and a precision of 0.999 suggests that out of the cases categorized as fake news, 99.9% were correct. These outcomes provide a strong indication that the BiLSTM model is highly proficient in the classification of fake and real news with a near-optimal overall performance based on the measures of recall and precision. It can be observed that the BiLSTM model achieves a higher accuracy score compared to the other models that have been discussed in this comparison including XGBoost. The confusion matrix gives a clear picture of the classification of fake news predicted by the XGBoost classifier.

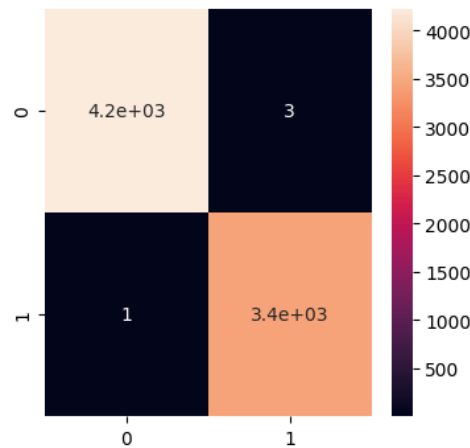


Figure 5.9: Confusion Matrix of BiLSTM classifier.

5.3 Results of Sentiment Analysis

Sentiment analysis is executed with the XGBoost model which had the best performance using only sentiment features for fake news detection. The results are illustrated in Table 5.7.

Performance Metrics	Score
Accuracy	73.973
Precision	65.134
Recall	74.209
F1-score	69.376

Table 5.7: Performance of XGBoost with sentiment features.

The model shows moderate performance but falls significantly short of its previous results with full feature set. With an accuracy of 73.973%, the model correctly classifies

about three-quarters of the news articles, which is notably lower. The precision of 65.134% indicates a higher false positive rate when compared to the full feature set. The recall of 74.209% and F1 score of 69.376% further confirms the model's reduced effectiveness. When using only sentiment features it is possible to achieve certain discriminative ability in the detection of fake news however it is not sufficient to achieve the same accuracy as models combining a wider range of features. This considerable performance reduction emphasizes the need to have complete and varied feature sets for fake news detection tasks.

Hence, another analysis was executed with XGBoost model using these sentiment features combined with already selected features. The results are illustrated in Table 5.8.

Performance Metrics	Score
Accuracy	98.103
Precision	97.600
Recall	98.197
F1-score	97.898

Table 5.8: Performance of XGBoost with all features.

The model performs much better when the sentimental features set is merged with previously chosen features. It displays outstanding results in each of the four evaluation metrics. This implies that both the underlying emotional tone and linguistic patterns are considered in this combination of features. This suggests that a well-chosen subset of features, including sentiment analysis, can be highly effective for fake news detection, potentially offering computational efficiency without significantly sacrificing accuracy.

5.4 Discussion of Research Questions

RQ1: Which machine learning model is most effective for predicting fake news and what metrics are most useful for evaluating its performance?

With regards to the performance of the machine learning models (Naive Bayes, SVM, Random Forest, and XGBoost), the XGBoost model demonstrated the highest efficiency in classifying fake news. It performed the best out of all four models based on the highest accuracy (98.181%) recorded for it. The XGBoost model's performance is followed

closely by SVM, then Random Forest, with Naive Bayes showing the least effectiveness. Out of all the metrics - accuracy, precision, recall or the F1 score - the F1 score, as well as the ROC curve, are most effective. However, F1 score gives balanced and effective evaluation of both precision and recall which are essential when false positives and false negatives affect the system. The AUC provides a view of how the model performs with respect to all possible classification thresholds. Thus, the high AUC proves that XGBoost is more accurate in classifying fake news from the real ones. Accuracy is quite straightforward and easy to understand while F1 score and AUC are more suitable in case when classes are unbalanced. The usage of these metrics gives a solid ground for the evaluation of the models in regard to fake news detection.

RQ2: Will the deep learning model outperform machine learning ensemble methods in terms of accuracy and efficiency in detecting fake news?

The Bidirectional Long Short-Term Memory (BiLSTM) model works better than the machine learning ensemble models, such as XGBoost in classifying the fake news. The BiLSTM model was extremely accurate with an accuracy of nearly 99.9%. These results are higher compared to the best performing ensemble model, XGboost, with an accuracy of 98.181%. As for the accuracy, it can be seen that the BiLSTM model has surpassed the ensemble models. The BiLSTM model, which has no false negative cases suggested by the 100% perfect recall score, means that it did not fail to identify any fake news. However, while using deep learning models such as BiLSTM there is a strong likelihood of resource intake and more time-consuming as opposed to the ensemble method. Thus, despite the fact that the BiLSTM model shows higher accuracy for identifying fake news, to give a more comprehensive comparison in terms of efficiency, more data such as training time, the inference speed, and resource consumption for each of the methods are needed.

RQ3: In what ways does sentiment analysis affect the machine learning model's overall efficacy in identifying fake news?

The impact of sentiment analysis on the proposed machine learning model is evaluated by comparing and contrasting the performance of the XGBoost model trained with only sentiment features to the model trained with additional features that were selected from the dataset. For the XGBoost model, the results were quite reasonable when only

sentiment features were selected with the accuracy of 73.973%. However, when the sentiment features are added along with selected other features, the model's performance dramatically improved, achieving an accuracy of 98.103%. These findings indicate that although the sentiment analysis gives a certain discriminative power in fake news detection, a higher accuracy is needed. Moderate accuracy with only sentiment features means that there is some link between the polarity of the articles and their credibility. However, the significant increase in the performance when sentiment is combined with other features shows that the sentiment is useful only when used in conjunction with other textual or contextual features. This way, the model can capture the sentiment aspect as well as the other language or structural features that fake news may contain and thus the model is more effective in detecting fake news. It is for these reasons that the results show that sentiment analysis does help in the detection of fake news but only when combined with other features.

Conclusions and Future Scope

6.1 Conclusion

For the fake news detection, the research has proven that the use of machine learning as well as deep learning was effective. The comparison of the performance of different models shows that although simple and fast traditional machine learning algorithms are used in many cases, they are most of the time surpassed in terms of accuracy by more complex deep learning models. In essence, the BiLSTM model proves to be most accurate, thus suggesting a superior capacity of the model to grasp distinctions inherent to the content of fake news. This explains why semantic information and content are critical facets in the detection process. The study's outcomes suggest that the inclusion of sentiment and contextual analysis features could provide a vast improvement to detection systems for fake news identifying multiple layers to provide a strong approach to dealing with such misinformation in digital media.

6.2 Future Scope

The future implications of this study involve the use of even more complex deep learning architectures like Transformer models and the application of attention mechanisms familiar in natural language processing. Besides, the integration of real-time data processing features might enhance the model's adaptability to the new fake news trends.

Another important area is the ability to deal with multilingual and multicultural information because disinformation is a global problem. Moreover, a detailed study of the effectiveness of different feature extraction techniques such as network-based features as well as User Behavior Analytics could prove useful. On the same note, including more detailed identification of emotions and psychological conditions in conjunction with sentiment analysis can also increase detection rates. Finally, constructing a comprehensive ensemble approach that combines the strengths of different models might be a much better way of handling the issue of fake news identification.

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